**PROGRAM**

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int main()
{
   char s[10][100], var[20];
   int val[10], n, i, j, k;
   printf("Enter number of statements: ");
   scanf("%d", &n);
   printf("Enter the statements:\n");
   for(i = 0; i < n; i++)
      scanf(" %[^\n]", s[i]);

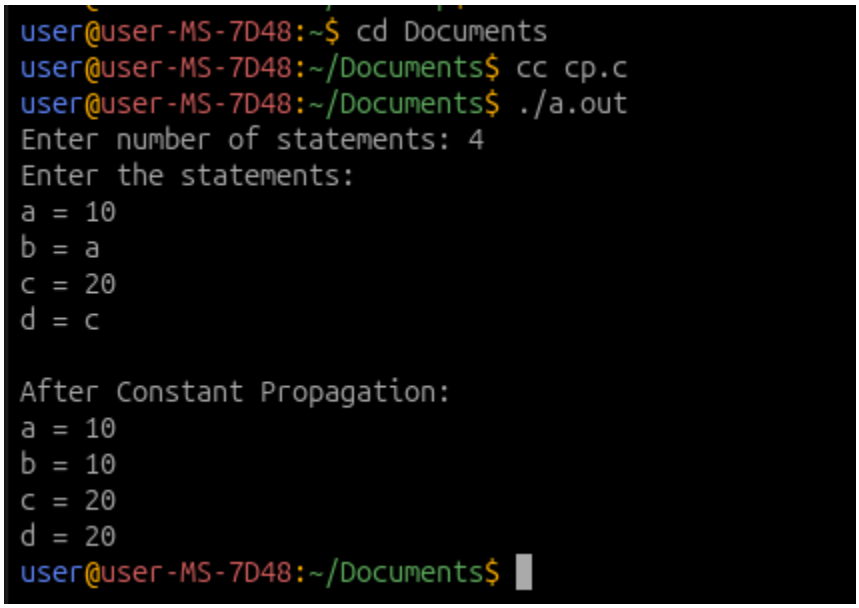
   printf("\nAfter Constant Propagation:\n");

   for(i = 0; i < n; i++)
   {
      for(j = 0; j < i; j++)
      {
         char name[20];
         int value;

         if(sscanf(s[j], "%s = %d", name, &value) == 2)
         {
            char *p = strstr(s[i], name);

            if(p != NULL && p != s[i] &&
               (p[-1] == ' ' || p[-1] == '='))
            {
               char temp[100];
               sprintf(temp, "%.*s%d%s",
                     (int)(p - s[i]), s[i], value,
                     p + strlen(name));
               strcpy(s[i], temp);
            }
         }
      }
      printf("%s\n", s[i]);
   }
   return 0;
}
```

OUTPUT


user@user-MS-7D48:~$ cd Documents
user@user-MS-7D48:~/Documents$ cc cp.c
user@user-MS-7D48:~/Documents$ ./a.out
Enter number of statements: 4
Enter the statements:
a = 10
b = a
c = 20
d = c
After Constant Propagation:
a = 10
b = 10
c = 20
d = 20
user@user-MS-7D48:~/Documents$